Clustering and Similarity Learning in Financial Markets: A Tutorial for the Practitioners

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Key Findings

1. Traditional peer-grouping methods, such as industry codes or static style boxes, are coarse and rigid. Modern investment requires **adaptive, data-driven similarity systems** that quantify closeness based on actual risk and thematic exposures, not pre-defined labels. Advances in metric learning and large language models enable the construction of flexible, operational neighborhoods for securities and funds.

2. Evaluation must shift from abstract academic metrics to operational validity. The core focus is on substitution fidelity, neighborhood stability across regimes, and segment utility for decision-making. This ensures that models are transparent, auditable, and withstand the scrutiny of compliance and fiduciary accountability.

3. Effective similarity requires moving beyond single data types to fuse heterogeneous sources. Models must integrate tabular fundamentals, time-series performance, textual disclosures, and relational graph structures. This allows practitioners to achieve robust peer discovery and align narratives with market dynamics, tackling the sparsity inherent in financial data.

Abstract

Clustering and similarity learning are increasingly indispensable for structuring heterogeneous financial data and supporting real-world decision making. Traditional heuristics such as industry codes, static style boxes, or return correlations, offer only coarse and rigid notions of peer groups. Recent advances in metric learning, graph methods, and large language models now make it possible to build adaptive neighborhoods of securities, funds, companies, and investors that align more closely with actual risk, liquidity, and thematic exposures. This tutorial synthesizes these methodological developments and demonstrates their use across major asset classes. Case studies show how supervised proximities improve bond substitution, how fund similarity systems reconcile category reproducibility with outlier detection, how multimodal pipelines refine company comparables for valuation and strategy, and how investor clustering enhances personalization and Know-Your-Client (KYC) analytics. We emphasize modeling choices that make clustering and similarity auditable and robust under regime shifts. We also outline their evaluation protocols such as neighborhood stability, substitution fidelity, and segment utility, etc., which align with investment, compliance, and fiduciary objectives. Overall, the central message for practitioners is pragmatic: similarity systems have moved beyond experimental prototypes and now stand as deployable techniques within real investment workflows.

Keywords: Clustering; Similarity Learning; Investment Management

Financial markets are built on comparisons. Every day, practitioners ask which bond can substitute for another, which fund belongs to which style, which company is closest to a competitor, or which investor segment shares similar behaviors. Clustering and similarity learning formalize these judgments, transforming them from ad hoc heuristics into systematic, data-driven frameworks. What began as rule-based taxonomies and simple ratio screens has grown into a toolkit that can integrate fundamentals, returns, text, graphs, and alternative data to uncover meaningful neighborhoods across diverse asset classes.

The promise of these methods is straightforward: they help organize complexity. Vast universes of securities, funds, or clients can be reduced to interpretable groups, and those groups can be aligned with real economic attributes such as credit risk, liquidity, or thematic exposure (Exhibit 1). Peer sets become the foundation for tasks as varied as pricing, hedging, allocation, surveillance, and compliance. Properly designed, clustering and similarity tools provide practitioners with a transparent way to align analytics with fiduciary and regulatory objectives.

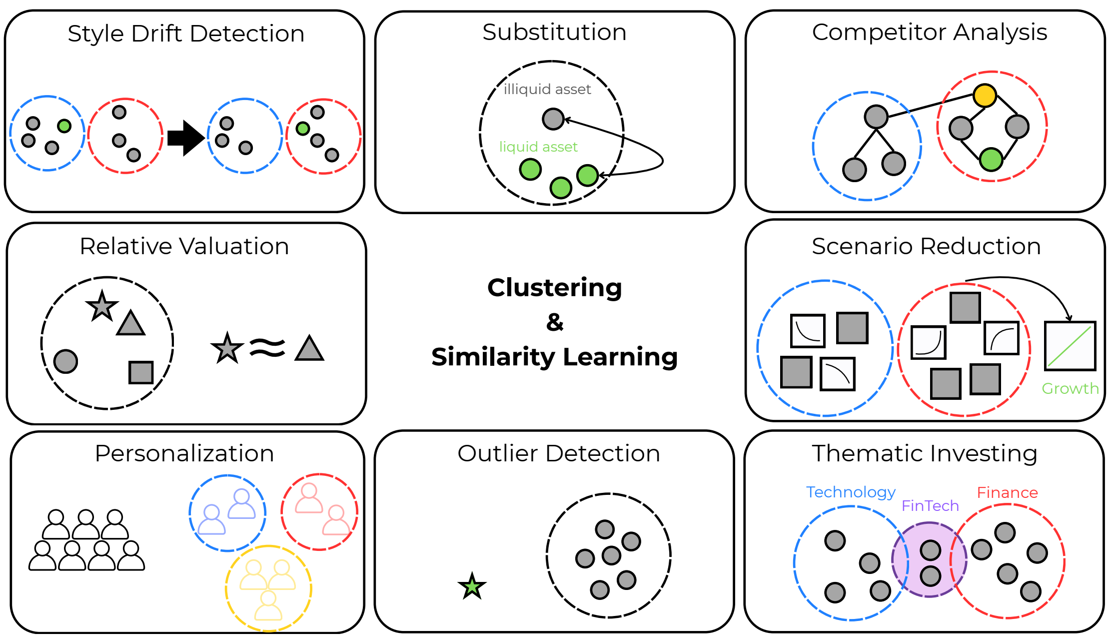

Exhibit 1: Usecases of Clustering and Similarity Learning in Financial Markets.

Yet practical use requires care. Unlike supervised prediction, similarity has no universal ground truth. The definition of "peer" depends on context, data modality, and market regime. Outputs can shift with parameter choices, and advanced models often sacrifice interpretability just when oversight is most needed. For this reason, governance frameworks such as SR 11-7 place strong emphasis on documentation, reproducibility, and transparency.

This tutorial delivers a pragmatic perspective. It reviews the range of clustering and similarity methods available today, shows how they operate across asset classes from bonds and funds to commodities and scenarios, and highlights evaluation practices that matter for production deployment. The aim is to equip practitioners with a field guide to design robust, interpretable, and scalable systems that make similarity learning a dependable part of financial workflows.

Clustering and Similarity Learning Methodologies

Before discussing different techniques, we first clarify important nuances of clustering and similarity learning. At a high level, *clustering* discovers groups directly from data without labels, while *similarity learning* focuses on constructing or learning a measure of closeness that can drive retrieval, ranking, substitution, or downstream modeling.

What practitioners mean by clustering

Clustering is the process of grouping financial instruments, funds, or investors into sets where members share similar characteristics, such as risk profile or performance behavior, while being different from those in other sets. Clustering is usually thought of as an unsupervised learning task, that is, the grouping is learned without resorting to any target variable(s). In this setting, the notion of closeness is typically governed by a pre-specified similarity or distance function (for example, Euclidean distance, correlation-based distance, or mixed-type dissimilarities), rather than being learned from labeled supervision (common distance metrics are summarized in Exhibit 2). Thus, the effectiveness of clustering hinges on how well this chosen metric captures the underlying economic structure of the data.

Clustering methods across modalities Clustering provides a flexible family of techniques to partition financial data into coherent groups, but the choice of algorithm depends on the modality, the available similarity measure, and the structure of the data. Classical clustering methods have a long tradition in finance (Marti et al., 2021), from peer group formation to portfolio construction, and remain central to many modern applications. Below, we describe the main approaches (summarized in Exhibit 3).

Partitioning methods such as k-means and k-medoids group securities by minimizing within-cluster distances (Jain et al., 1999). They are fast and effective for tabular data in equities and bonds, though they require a pre-set number of clusters; k-medoids is more robust to outliers.

Hierarchical clustering builds nested trees. Dendrograms enable drill-down views and align with portfolio hierarchies. In practice, it supports Hierarchical Risk Parity (Lopez de Prado, 2016), which delivers more stable allocations under estimation error (Raffinot, 2017).

Density-based clustering (DBSCAN, HDBSCAN) defines clusters as dense regions separated by sparse areas. It avoids pre-specifying cluster counts and is well-suited to detecting niche cohorts or anomalies such as fraud and rare assets.

Model-based clustering assumes data come from a mixture of distributions, often Gaussian. It yields soft memberships and provides natural uncertainty estimates.

Finance often faces *mixed-type tabular data* that combinine numeric, ordinal, categorical, and sparse indicators. Measures such as Gower (Gower, 1971), k-prototypes, and generalized dissimilarities allow clustering without forcing all variables into numeric space.

Time-series clustering extends these methods to sequential data. Distances include correlation, ultrametrics from minimum spanning trees, dynamic time warping (DTW), Kullback–Leibler divergence, Wasserstein distance, and others. Rolling windows reveal co-movement regimes while adapting to structural breaks.

Text clustering embeds documents via TF–IDF or transformer models, then applies partitioning or hierarchical methods (Manning et al., 2008). Applications include grouping prospectuses, earnings calls, and filings to integrate qualitative signals.

Graph clustering analyzes relational data, where nodes represent entities and edges represent economic ties. Methods include community detection, modularity optimization, and embeddings such as node2vec (Grover & Leskovec, 2016; Satone et al., 2021), widely used in systemic risk and peer discovery.

Image and spatial clustering uses embeddings from convolutional or Siamese networks with partitioning or density-based grouping. In real estate, property and satellite images are clustered for comparables, collateral risk, and geographic exposure (Poursaeed et al., 2017).

What practitioners mean by similarity learning

Whereas typically clustering methods take a predefined notion of "closeness" in the form of a prechosen similarity or distance metric, similarity learning formalizes the notion of closeness by *learning* the distance metric using the given data: the goal here is to learn a distance or similarity function that keeps "positives" close and "negatives" apart (contrastive and triplet paradigms), with extensive surveys covering both classic vector data and structured objects (Kulis, 2013).

In practice, this takes several forms. *Supervised similarity* uses known equivalences—such as funds in the same category or bonds with interchangeable features—to shape embeddings, with product taxonomies or analyst coverage serving as weak labels. *Semi-supervised and self-supervised similarity* leverage abundant unlabeled data, for example by forecasting in time series or masked modeling in text, reducing dependence on noisy human labels.

Exhibit 2: Common Distance Metrics for Clustering by Data Type

Data Type	Distance / Similarity Metrics	Notes / Usage
Numeric (continuous)	Euclidean Manhattan (L1) Minkowski Mahalanobis	Standard metrics for tabular and multivariate data; Mahalanobis accounts for correlations
Categorical	Hamming distance Simple matching coefficient Jaccard similarity	Works for binary or categorical variables; Jaccard ignores co-absences
Mixed-type (numeric + categorical)	Gower's similarity Huang distance Kernel metric dissimilarity	Handles heterogeneous attributes without coercing everything to numeric
Time series	Correlation-based distance Dynamic Time Warping (DTW) Euclidean on lagged vectors Wasserstein distance	Captures similarity in trajectories, alignment, and distributional shifts
Text	Cosine similarity on TF–IDF Word2Vec or BERT embeddings Jaccard similarity on token sets	Measures semantic similarity between documents, disclosures, or strategies
Graphs / Networks	Shortest-path distance Graph edit distance Spectral distance Node2Vec / graph embeddings	Used for issuer–security networks, fund–holding bipartite graphs
Images / Spatial data	Euclidean on feature vectors Structural similarity (SSIM) Siamese embedding distances	Property or satellite image clustering; spatial proximity also relevant
Distributional	Kullback–Leibler divergence Jensen–Shannon divergence Wasserstein (Earth Mover's) distance	Comparing empirical return distributions, volatility regimes, or scenarios

Another distinction is between *global* metrics, which apply across all entities, and *local* or adaptive metrics, which adjust by region, class, or query. Finance often requires *task-specific similarity* (for example, liquidity substitution versus credit risk), supported by multi-task learning that shares information while tailoring metrics to objectives. Finally, feature-weighted similarity methods emphasize the attributes most relevant to the task, ensuring neighborhoods and clusters reflect real operational needs (Ghashti & Thompson, 2025a,b).

In practice, this enables models to focus on the features most relevant to the financial task at hand, whether prioritizing liquidity, credit, temporal dynamics, or textual content, thereby producing neighborhoods and clusters that better reflect operational realities.

Exhibit 3: Clustering methods across data modalities in finance

Modality	**Clustering Method**	**Notes / Usage**
Tabular data	k-means / k-medoids	Partitions data to minimize intra-cluster variance; used for fundamentals, factor exposures.
	Hierarchical clustering	Creates a hierarchy of clusters (dendrogram); applied to equities, funds, macro data.
Tabular / mixed data	Density-based (DBSCAN, HDBSCAN)	Finds dense regions, good for irregular shapes and noise; used for transaction fraud, anomaly detection.
Tabular / distributions	Model-based clustering (Gaussian mixtures)	Probabilistic approach for soft assignments; models heterogeneity in returns or volatility regimes.
Time series	Correlation clustering Dynamic time warping Spectral clustering	Groups assets by co-movement or trajectory similarity; for portfolio diversification, regime analysis.
Text	Document embedding + k-means/hierarchical clustering	Clusters documents based on semantic embeddings; thematic analysis of filings, reports.
Graphs	Community detection Spectral clustering Modularity optimization	Finds dense subgraphs (communities); analyzes interbank networks, fund–holding structures.
Images / spatial	CNN / Siamese embeddings + clustering	Clusters based on learned visual features; used for real estate comps, satellite data analysis.

Similarity learning in finance goes beyond embedding vectors into latent spaces. It often requires *structured similarity*, where relationships such as portfolio proximities, issuer–security hierarchies, or temporal links are preserved. Graph-based and multimodal methods capture these structures, ensuring that closeness respects domain constraints. Overall, similarity learning is not a single technique but a flexible toolkit, adaptable to the context-specific definitions of "peer" that practitioners need.

Similarity learning methods Modern finance provides diverse modalities such as tabular, time series, text, images, and graphs, and so on, and similarity methods must adapt accordingly. Below we outline the main families of methods (summarized in Exhibit 4).

Metric learning (Kulis, 2013) transforms input space so that similar instances move closer and dissimilar ones farther apart. Classical methods include Mahalanobis and LMNN, which learn linear transformations to reshape geometry around task-specific closeness. Kernel metric learning extends this to heterogeneous data by operating in reproducing kernel Hilbert spaces, handling mixed variables and nonlinearities for applications such as segmentation and credit cohorts.

Random forest proximity measures how often two observations fall into the same leaf across an ensemble, naturally handling nonlinearities and mixed data (Breiman, 2001). It has been applied to bond substitution, fund peer grouping, and investor segmentation (Jeyapaulraj et al., 2022; Saha et al., 2024), where supervised proximities align similarity with predictive tasks.

Siamese networks map inputs into embedding spaces using paired examples with contrastive loss (Bromley et al., 1993). They support real estate comparables, document matching, and collateral risk assessment (Poursaeed et al., 2017; Das et al., 2021; Viana & Barbosa, 2021). *Triplet loss embeddings* extend this by enforcing anchor–positive–negative ordering, enabling fine-grained similarity for thematic investing (Schroff et al., 2015; Lee et al., 2025).

Time-series similarity learning uses correlation, DTW, or Wasserstein distances, with recent encoders adapting to regime shifts (Dolphin et al., 2023).

Textual similarity relies on transformer embeddings such as BERT (Devlin et al., 2019), clustering disclosures and reports to support thematic analysis and peer discovery (Vamvourellis et al., 2022; Gabaix et al., 2024; Vamvourellis et al., 2024).

Graph neural networks (GNNs) learn node embeddings from neighborhood structure, useful for issuer–security links, fund–holding networks, and supply chains. Methods like DeepWalk (Perozzi et al., 2014) and node2vec (Satone et al., 2021) support systemic risk monitoring and peer retrieval.

Contrastive learning optimizes positive vs. negative pairs rather than triplets (Chen et al., 2020), and is applied in time-series representation and volatility regime learning (Dolphin et al., 2024).

Multimodal similarity integrates fundamentals, returns, text, and graphs into joint embeddings, using transformers and attention to align signals. Examples include theme grouping (Lee et al., 2025) and ripple-effect prediction via LLM–graph integration (Xu et al., 2025).

Open-set similarity adds anomaly detection, flagging observations outside learned groups to identify new strategies, anomalies, or risk clusters (Feng et al., 2024).

Exhibit 4: Similarity Learning Methods (Supervised and Semi-supervised)

Method	Learning Type	Data Type / Input	Applications
Metric learning	Supervised	Tabular features with labels or expert constraints	Bond substitution Fund peer groups Investor risk profiling
Kernel metric learning	Supervised / Semi-supervised	Mixed-type tabular variables with side information	Household finance segmentation KYC similarity
Siamese networks	Supervised (pairwise labels)	Images embeddings Text embeddings Tabular embeddings	Real estate comparables Document similarity Collateral risk
Graph neural networks	Supervised / Semi-supervised	Networks (issuer–security fund–holding)	Systemic risk monitoring Fund embeddings Corporate peer sets
Contrastive learning		Time series embeddings Text embeddings Multimodal inputs	Regime-aware asset Thematic investing Factor neighborhood Fund classification Disclosure similarity
Multimodal similarity learning		Integrated inputs (tabular, text, image, graph)	Company comparables Fund style innovation ESG alignment
Open-set similarity learning	Supervised with anomaly detection	Tabular + text embeddings	Novel fund styles Outlier detection Market anomaly identification

Evaluation Methodologies in Clustering and Similarity Learning

Evaluating clustering and similarity is difficult because most methods are unsupervised. In finance, the focus has shifted from abstract metrics to *operational validity*: whether peer groups remain stable, substitutes preserve liquidity and risk, outliers align with expert judgment, and outputs are explainable for compliance. Models are judged by their ability to support investment decisions, withstand regulatory scrutiny, and meet fiduciary duties.

Academic metrics. Classical indices such as Silhouette, Dunn, and Davies–Bouldin measure cohesion and separation, while external measures like ARI and NMI test alignment with known labels, often in fund category reproducibility (Castellanos et al., 2024). Stability across bootstrap samples or rolling windows is especially important in equities and ETFs (Polinesi & Recchioni, 2021).

Practitioner validation. Bonds are tested via K-nearest neighbor (KNN) peer sets that preserve credit and liquidity (Jeyapaulraj et al., 2022). For funds, models must reproduce vendor categories such as Morningstar (Mehta et al., 2020; Desai & Mehta, 2021; Feng et al., 2024). In equities, peer groups should explain return co-movement better than GICS or SIC codes (Barberis et al., 2005; Molinari et al., 2024). For investors, utility is judged by improved segmentation, personalized advice, or behavioral monitoring (Thompson et al., 2021; Thompson & Davison, 2024).

Multimodal robustness. As data sources expand, evaluation extends to multimodal pipelines. Text-based peers are validated by valuation accuracy gains (Dzuyo et al., 2025), graph embeddings by improvements in risk and counterparty monitoring (Zhou et al., 2022; Cao et al., 2024), and LLM-driven systems by ablation tests and governance checks such as bias audits (Papenkov et al., 2023).

Exhibit 5 provides a brief summary of various evaluation metrics.

Exhibit 5: Evaluation methodologies: academic vs. practitioner perspectives.

Domain	**Academic metrics**	**Practitioner metrics**
General clustering	Silhouette, Dunn, Davies–Bouldin, ARI, NMI (Hubert & Arabie, 1985; Jain et al., 1999; Castellanos et al., 2024)	Stability in rolling windows; interpretability for compliance
Bonds	Cluster cohesion vs. labels	Substitution fidelity under liquidity and credit constraints (Jeyapaulraj et al., 2022; Rosaler et al., 2025)
Funds	ARI/NMI vs. vendor labels	Category reproducibility, outlier detection, open-set anomalies (Mehta et al., 2020; Desai & Mehta, 2021; Feng et al., 2024)
Equities	Return correlation alignment with GICS/SIC	Thematic co-movement, factor exposure stability (Barberis et al., 2005; Molinari et al., 2024)
Investors	Clustering indices on transaction data	Segment utility for KYC, personalization, market crash behavior, monitoring (Thompson et al., 2021; Thompson & Davison, 2024)
Multimodal	Cross-modal embedding consistency	Contribution of each modality, ablation robustness, governance audits (Dzuyo et al., 2025; **?**)

Fixed-Income Asset Clustering and Similarity

Fixed income poses unique challenges for clustering and similarity (Fabozzi, 2007). Bonds differ widely in covenants, maturity, callability, and tax status, while trading is often sparse and noisy. Investor needs center on substitution (finding liquid peers), relative value (benchmarking within a cohort), and risk diagnostics (flagging outliers or monitoring). Therefore, effective pipelines combine tabular descriptors such as ratings and fundamentals with time-series behavior, textual disclosures, and graph structures like issuer linkages or co-holdings to create peer groups that are both stable and responsive to market regimes.

Corporate Bonds

In corporate credit, the most valuable definition of "similar" is one for trades: a substitute must preserve exposures such as sector, rating, maturity, and optionality, while also working under real frictions like inventory, lot size, and liquidity. Because trading is intermittent and spreads embed liquidity premia, desks need peer cohorts to price, hedge, and source alternatives when a target CUSIP is unavailable.

Unsupervised clustering can identify peer sets based on descriptive features, but its value depends on business-centric validation such as substitution under desk constraints, turnover, and rolling stability, i.e., not on abstract indices. Wu demonstrates this by generating corporate bond cohorts under rating, sector, and duration priors, then validating them against desk needs (Wu, 2023).

Supervised similarity provides a more direct path. Jeyapaulraj et al. show how Random Forest proximities learn bond–bond distances aligned with desk use cases such as relative value, substitution, and surveillance, and propose evaluation criteria based on stability and substitution fidelity (Jeyapaulraj et al., 2022). Rosaler et al. extend this idea with a quantum-cognition metric learner that improves on tree models in high-yield markets, where noise and sparsity challenge heuristic similarity (Rosaler et al., 2025).

Relational structure adds another layer. Bonds co-move not only through shared terms but also through issuers, capital structures, and common ownership. Graph methods capture these links. Zhou et al. use a temporal bipartite fund–bond graph to show that graph neural networks improve price and yield prediction relative to time-series models, with node embeddings naturally supporting peer retrieval (Zhou et al., 2022). Similar graph-based models for default risk confirm the value of encoding issuer and ownership networks (Li et al., 2023).

Portfolio trading further underscores the need for robust similarity. Nearly one-fifth of U.S. investment-grade volume now trades in baskets. Traditional overlap metrics often fail to align baskets with ETF benchmarks. STRAPSim addresses this by matching bonds at the constituent and weight level, producing closer return-based alignment and improving pricing and execution in portfolio trades (Li et al., 2025). For practitioners, basket similarity enhances transparency, liquidity sourcing, and efficiency in less liquid credit markets.

Municipal bonds

Municipal markets add another layer of complexity: issuers range from states to school districts, tax treatments vary, and covenants are highly idiosyncratic. Trading is sparse, dealer-intermediated, and costly, with intermittent price discovery (Harris & Piwowar, 2006; Green et al., 2007). In this setting, supervised similarity has shown strong value. A CatBoost-based framework learns muni–muni distances from large U.S. datasets and assembles nearest-neighbor cohorts for pricing and surveillance, outperforming heuristic rules in backtests (Saha et al., 2024). This aligns with desk workflows such as peer price checks conditioned on risk, tax status, and structure, monitoring category drift, and identifying comparable trades.

Similarity in munis goes beyond tabular terms like coupon, maturity, or ratings. Textual disclosures in official statements add insight on pledges and covenants, and their complexity is itself priced (Farrell et al., 2023). Relational signals such as co-holdings, issuer hierarchies, and geography help stabilize cohorts when trading is sparse. Evaluation emphasizes operational validity: stability over rolling windows, substitutes under liquidity constraints, and agreement with expert judgment. Marketwide municipal securities statistics[1] provide baselines for coverage, turnover, and regime shifts.

Securitized credit: MBS, CMBS, ABS

In Mortgage-Backed Securities (MBS) and Asset-Backed Securities (ABS), substitution and risk are contingent upon borrower behavior, particularly prepayments and defaults. Classical models segment pools by loan and macro characteristics such as coupon, seasoning, and geography, and estimate cohort-level termination hazards (Kang & Zenios, 1992). Bayesian approaches model prepayment at the pool level with hierarchical mixtures, capturing path dependence and unobserved heterogeneity.

In commercial Mortgage-Baked Securities (CMBS), the key dimension of similarity is termination risk, since prepayment and default drive pricing and surveillance. Early loan-level hazard models provided a natural basis for peer cohorts by segmenting mortgages with comparable termination propensities (Ambrose & Sanders, 2003). Mixture survival methods sharpen these cohorts by distinguishing long-term survivors from at-risk loans (Yildirim, 2008). Geography also matters: local economic conditions such as property values, unemployment, and house price appreciation significantly improve default modeling and should be built into similarity neighborhoods (An et al., 2013). Evidence from special-serviced loans shows that workout strategies and default probabilities depend on market conditions, supporting grouping stressed loans by resolution path (Chen & Deng, 2013). Pricing also reflects information frictions: origination channel (conduit vs. portfolio) introduces systematic differ-

[1] https://www.msrb.org/

ences consistent with adverse selection, which means similarity models for pricing must account for data provenance (An et al., 2011). Taken together, effective CMBS peer sets rely on hazard-consistent features such as contract terms, collateral, geography, macro drivers, and origination channel, rather than surface descriptors.

Housing and collateral markets (linkages to fixed income)

House price and collateral risk models increasingly rely on representation learning from spatial, textual, and visual data. Vision and multimodal embeddings that combine property photos with street and satellite context improve valuation and comparables search relative to traditional methods (Poursaeed et al., 2017; Law et al., 2019; Nouriani & Lemke, 2022). At scale, Siamese networks learn "similar home" embeddings that drive recommendations and peer retrieval, showing how task-aligned similarity supports comparables in live marketplaces[2]. Graph and spatial embeddings extend this further by incorporating neighborhood amenities and geographic context. For example, multipartite graphs that link homes to nearby points of interest improve prediction accuracy, while attention-based spatial interpolation weights neighbors according to learned relevance. Both approaches strengthen comparables and provide robust stress testing for collateralized products (Das et al., 2021; Viana & Barbosa, 2021).

Fund Clustering and Similarity

Peer grouping is central to fund analysis. Allocators, such as multi-asset managers and consultants, use peer groups to construct balanced portfolios, test whether exposures are redundant or complementary, and design allocations that deliver real diversification. Fund selectors on advisory platforms rely on peer groups to find comparables and to spot outliers that drift from their stated style or category, which can reveal opportunities or hidden risks. Sales and product teams apply similarity to identify competitor funds that can be substituted with home-grown products on recommended lists, a process that requires precise and auditable measures of substitutability. Regulators and oversight bodies also use peer analysis to check that funds are marketed consistently with their mandates and that holdings and performance align with disclosures.

Across these use cases, the peer group is a critical tool. It guides allocation and diversification, supports distribution and competitive positioning, diagnoses style drift or innovation, and enforces compliance with disclosure standards.

[2] https://www.zillow.com/tech/embedding-similar-home-recommendation/

Categorizations, Clustering and Similarity for Funds

ETFs are often analyzed alongside mutual funds in peer-group pipelines. Early work relied on return-based style analysis, with Sharpe's method providing a disciplined way to decompose returns into style exposures and guide category design (Sharpe, 1992). Marathe & Shawky (1999) documented misclassifications in vendor taxonomies by applying return-based clustering, starting with K-means and related methods. Similar findings emerged in studies of equity and mutual fund categories (Kim et al., 2000).

Through the 2000s and 2010s, clustering was used to uncover structure beyond static labels. Nonlinear methods such as self-organizing maps refined or challenged official categories, as shown in Spanish mutual funds (Moreno et al., 2006). Double-stage clustering compressed performance and risk measures before grouping them into interpretable cohorts (Menardi & Lisi, 2015). Flori et al. (2021) shifted from returns to holdings, capturing similarity through portfolio commonality. Network views of fund–asset bipartite graphs revealed how popular versus niche holdings relate to performance and crowding risk.

For years, unsupervised clustering was used as a benchmark against vendor labels like Morningstar[3] and Lipper[4], with the assumption that disagreement implied flawed categories. Desai & Mehta (2021) demonstrated that this was misleading, as clustering outcomes depend heavily on feature choice, distance metrics, and hyperparameters; thus, divergence often reflects algorithmic bias rather than label failure. Castellanos et al. (2024) and Desai et al. (2023) reinforced this by stressing the role of evaluation design. To resolve the issue, supervised validation frameworks have demonstrated that categories, such as those used by Morningstar, are reproducible with fund features, proving they are systematic rather than arbitrary (Mehta et al., 2020). Later work applied metric learning and supervised similarity to extend these taxonomies, aligning distance functions with declared categories and flagging inliers and outliers in a manner that is transparent to practitioners and regulators (Desai & Mehta, 2021; Desai et al., 2023).

Recent work expands fund similarity in three directions. DeMiguel et al. (2023) showed that supervised models on fund characteristics can identify managers with persistent positive alpha, highlighting predictive modeling as a complement to similarity analysis. Martín-Melero et al. (2025) compared sector involvement with financial features for ESG scoring, finding that tree-ensemble methods perform robustly across modalities. Zou & You (2025) proposed an integrated framework that clusters funds by performance patterns and then tests group persistence, outperforming style-box baselines and producing persistence-aware peer sets for monitoring.

Supervised similarity frameworks bring these elements together by quantifying how far a fund is

[3] https://advisor.morningstar.com/Enterprise/VTC/MorningstarCategoryClassificationforFunds_October2024.pdf

[4] https://lipperalpha.refinitiv.com/lipper/fund-classifications/

from its declared category and flagging outliers with diagnostics that are transparent to selectors and oversight teams (Desai et al., 2023). Recent work extends this approach with open-set recognition. For example, Random Forests can be enhanced with distance-based rules that both classify funds into known categories and detect when a fund lies outside all recognized clusters (Feng et al., 2024).

Similarity also strengthens explainability. Yampolsky et al. (2024) introduced case-based methods for Random Forests that highlight prototypes, critics, counterfactuals, and semifactuals from training data to clarify why a model produced a given outcome. Applied to funds, this means that similarity tools can retrieve nearest peers, surface representative exemplars, and flag boundary cases, offering interpretable diagnostics that help selectors and auditors trust and act on the results.

Clustering and Similarity of Funds Based on Returns

In funds and ETFs, return histories are a practical basis for peer discovery and style checks. Early work clustered return paths into interpretable exposures, creating peer groups that could be maintained with minimal labeling (Pattarin et al., 2004). Subsequent studies linked fund uniqueness, measured relative to time-series neighbors, to flows, fees, and alpha persistence (Vozlyublennaia & Wu, 2016). For ETFs, distance choice matters—Euclidean, DTW, and shape-preserving methods produce different clusters, especially during shocks like COVID, and influence portfolio construction (Ito et al., 2021).

Noise-reduction techniques, such as correlation-matrix filtering, help recover ETF groupings consistent with taxonomies (Polinesi & Recchioni, 2021). Returns-plus-risk approaches add volatility states to the model, yielding peer sets that capture co-movement and regime-sensitive risk (Lisi & Otranto, 2008).

Clustering and Similarity based on other data modalities

Beyond flat labels, graph embeddings of fund–holding networks such as node2vec generate Fund2Vec representations that capture relational structure, supporting peer search, recommender systems, and competitor analysis (Satone et al., 2021). Related work adapts language models like Word2Vec and BERT to create asset embeddings from portfolio structures, which improve the prediction of valuations, co-movement, and trading activity (Gabaix et al., 2024). Text-based methods also contribute: NLP on prospectuses and strategy descriptions reproduces category systems and produces semantic fund embeddings that align with practitioner views of peers (Vamvourellis et al., 2022).

Table 7 summarizes clustering and similarity methods and their applications for funds.

Company Clustering and Similarity

Identifying comparable companies supports valuation, risk management, strategy, and sales. Analysts use peers for relative valuation by applying multiples, while risk teams and portfolio managers rely on peer groups to diversify, hedge, and monitor exposures. Corporate strategy teams benchmark competitors and map markets, Mergers & Acquisitions (M&A) professionals search for targets with similar profiles, and commercial or underwriting desks use peers for lead generation and risk assessment. Robust similarity measures, therefore, strengthen decision-making by aligning company neighborhoods with the attributes that matter in each workflow.

Traditional taxonomies such as SIC (Guenther & Rosman, 1994) and GICS[5] offer a baseline for defining peers, but they are coarse and static. Firms sharing the same industry code can differ substantially in terms of products, customers, or technology, and such labels do not produce ranked peer lists necessary for comparables and surveillance. These shortcomings motivate data-driven approaches that integrate fundamentals, time-series signals, text, graphs, and alternative data into continuous similarity scores and flexible clusters for both public and private firms.

Traditional approaches and their limits

Historically, practitioners supplemented industry codes with financial-ratio comparisons and simple clustering. (Bhojraj & Lee, 2002) showed that selecting peers from fundamentals rather than labels improved screening and explanatory power. Classification from financial statements alone can recover sectors with reasonable accuracy using tree or boosting models (van der Heijden, 2022). Meanwhile, network heuristics, such as "common-analyst" coverage, also signal relatedness across firms (Kaustia & Rantala, 2021). These methods were useful but limited, as they ignore qualitative aspects of business models and lack the multimodality and adaptivity now required for large-scale peer analysis.

Modern data-driven approaches by modality

Machine learning (ML) on fundamentals supports goal-specific peer sets and clustering. Correlation-based methods have been used to form tradable portfolios for asset pricing (Ahn et al., 2009) and to classify stocks into groups that maximize Sharpe ratios in mean-variance frameworks (Kim et al., 2014). Other work applies K-medians or metric learning on financial ratios (Cai et al., 2016; Dzuba & Krylov, 2021), and designs "financial clusters" from statement structures that sharpen explanations beyond industry codes (Fodor et al., 2021). Supervised relative valuation methods use ML to express

[5] https://www.msci.com/our-solutions/indexes/gics

predicted multiples as peer-weighted averages, creating valuation kernels that are interpretable and aligned with comparables (Geertsema & Lu, 2023).

On the time-series side, similarity is often judged by return co-movement. Equity2Vec links news co-mentions, technical indicators, and sentiment to capture evolving cross-company relations, improving prediction accuracy and trading profitability (Wu et al., 2021). SimStock tackles market non-stationarity by treating time periods as separate domains and modeling parameter drift, blending time-series features with firm fundamentals (Hwang et al., 2023a). Other approaches demonstrate that sparse-autoencoder text embeddings improve correlation alignment compared to standard taxonomies (Molinari et al., 2024), while evolving probabilistic industry mixtures outperform GICS in predicting correlations (Papenkov, 2024).

Multimodal pipelines are also emerging. CUBAN combines textual descriptions with peer financials to forecast revenue and profitability, highlighting how comparables-informed features can enhance predictive performance (Bae & Hyde, 2024).

Textual data (NLP-based similarity). Text offers a way to capture product–market proximity and competitive positioning beyond numeric features. Early work showed that cosine similarity of 10-K descriptions reveals competitors, supports M&A analysis, and builds evolving industry networks (Hoberg & Phillips, 2010). Modern LLM-based methods embed company descriptions to reproduce GICS-like structures while enabling continuous peer ranking and performance alignment (Vamvourellis et al., 2024). Benchmarks across transformers and classifiers show strong sector classification from text alone, with LLM-generated augmentation improving robustness (Rizinski et al., 2024). Hybrid approaches that link narratives to accounting features boost both accuracy and interpretability in sector prediction (Dzuyo et al., 2025).

Beyond filings, unstructured website text and images can be embedded for peer identification and alternative industry segmentation, outperforming traditional NACE codes (Gerling, 2024). The THEME framework aligns firm profiles with thematic descriptions and return dynamics, producing adaptive embeddings for theme-based investing (Lee et al., 2025). Multi-task BERT fine-tuning on annual reports has also been used to jointly learn sector classification, return similarity, and sector naming (Ito et al., 2020).

Governance needs drive demand for interpretable similarity. Sparse autoencoders can disentangle LLM embeddings into transparent factors linked to fundamentals and returns (Molinari et al., 2024). Probabilistic multi-industry models further allow firms to load on several evolving topics, reflecting conglomeration and industry change (Vamvourellis et al., 2024; Papenkov et al., 2023; Papenkov, 2024). Finally, text-driven retrieval powers interactive peer-search systems that combine LLM reading, API queries, and ranking for live company lists (Patil et al., 2025).

Graph and network data. Relational data, such as supplier–customer ties, partnerships, shared investors, and board interlocks, capture economic proximity that fundamentals or text may overlook. CompanyKG builds a large-scale knowledge graph with heterogeneous relations and node text features, providing benchmark tasks for similarity prediction and competitor retrieval (Cao et al., 2024). Results show the value of combining node, edge, and hybrid information to quantify peers. Graph methods such as GNNs and link-prediction models learn embeddings that reflect multi-hop relationships and complement NLP by re-ranking text-based candidates with relational evidence. Earlier heuristics like common-analyst coverage also fit naturally as edges in this broader graph view (Kaustia & Rantala, 2021).

Other studies model the correlation matrix itself as a graph. Sarmah et al. show that learning node embeddings from such graphs maps correlated stocks into a low-dimensional space, enabling peer discovery, portfolio diversification, and systemic risk monitoring (Sarmah et al., 2024).

Images and alternative data. Visual cues (product imagery, brand style) and digital traces (web and app usage) can provide orthogonal signals of market positioning, especially in consumer-facing domains. Emerging work shows image embeddings can retrieve plausible peers and complement text/graph views (Kaczmarek & Pukthuanthong, 2024). While not yet mainstream in academic literature, these features increasingly appear in proprietary pipelines and can refine neighborhoods when combined with fundamentals, text, and graphs.

Private company similarities. Private company similarity is of growing importance for valuation, underwriting, venture capital, and corporate strategy. *Private-to-private* similarity supports tasks such as benchmarking financial performance among financial analysts, detecting high-growth firms, screening for credit risk cohorts, and mapping competitive landscapes in opaque markets. *Private-to-public* similarity enables anchoring startups to listed peers for relative valuation, benchmarking of disclosure and ESG alignment, and constructing synthetic price or risk proxies where no market data exist. Together, these workflows extend the reach of clustering and similarity learning from liquid, information-rich public issuers into domains where traditional comparables are sparse but decision-making still demands peer sets.

A first line of research leverages *structured registry data* such as Bureau van Dijk's *Orbis* to build financial panels of private firms. Studies show that financial-ratio clustering can recover sector structures and support benchmarking of profitability, leverage, and cash–flow patterns at scale (Kalemli-Özcan et al., 2024). Time-series analyses of private firms' employment or sales trajectories classify growth paths (for example, rapid-growth "gazelles" versus stable incumbents) via trajectory-aware clustering (Delmar et al., 2003; Coad, 2009). These approaches ground peer discovery in accounting and real-economy signals where coverage is sufficient.

When structured financials are missing, *textual data* become the dominant modality. Company descriptions, website blurbs, and profile summaries are encoded into embeddings to align startups with both private and public peers. For example, Company2Vec produces text-based embeddings of German companies for competitor retrieval (Gerling, 2024), while topic models and transformer embeddings map global startup ecosystems (Savin et al., 2023). These text-driven methods enable semantic comparables where numeric data are sparse.

Graph and alternative data extend coverage further. Large-scale heterogeneous knowledge graphs, such as CompanyKG (Cao et al., 2024), integrate millions of public and private firms with relational edges that span investors, board members, partnerships, and co-mentions. Graph neural networks and link-prediction models learn embeddings that more accurately recover competitors and partners than text alone. Complementary signals from imagery and digital traces also emerge: product photos, brand cues, and app usage data can position consumer-facing firms semantically, enriching private-to-private comparables (Kaczmarek & Pukthuanthong, 2024). Together, these strands show that multimodal similarity systems can extend valuation, strategy, and risk tools into the opaque world of private firms, supporting both intra-private benchmarking and mappings onto public market anchors.

Investors Clustering and Similarity

Worldwide, financial institutions are tasked with understanding their client base. Based on this understanding, these institutions recommend several assets and monitor the performance of client portfolios. They are often required to update their understanding of clients as their decision-making changes over time, as investment needs, including financial objectives and risk tolerance change as clients age and life situations change. This understanding can be gained by measuring investor risk tolerance (Thompson et al., 2022) and risk aversion (Grable et al., 2024) relative to portfolios through statistical quantities. Similarity learning offers an alternative approach to measuring clients' risk by understanding clients as different investor types and their relation to an institution's overall portfolio.

Investor decision-making and risk assessment frameworks

Before discussing similarity learning for assessing investor risk and suitability, we first clarify how clients make investment decisions and how these decisions can be mapped to regulatory and practical frameworks. Investor profiling seeks to directly elicit preferences, risk tolerance, and time horizons from clients, while revealed behavioral analysis focuses on comparing these elicited preferences with actual investment choices (Faff et al., 2004).

How individuals make investment decisions, and particularly how they understand, tolerate, and manage financial risk, has been a central concern in regulatory practice. At a high level, regulators, such as the Financial Industry Regulatory Authority[6] (FINRA) of the USA and the Canadian Investment Regulatory Organization[7] (CIRO) of Canada, develop suitability regulations for securities-related commerce to protect investors. They require collecting client information, including identity, investment needs, financial objectives, and circumstances, as well as risk tolerance, to ensure that recommended products and behaviors align with client financial goals. In practice, firms implement suitability largely through onboarding processes, such as questionnaires and demographic screenings. Classical risk assessment approaches have been used in financial institutions for decades. Suitability assessments often use questionnaire-based methods, risk assessment tools, and asset calculators, among others, which can then be used to recommend suitable assets for portfolios Baloyi & Lotter (2024) and inform risk tolerance scales Thompson et al. (2022), thereby grouping clients into broad risk categories.

A central question in academic research surrounding suitability is whether current investor risk assessments meaningfully reflect behavioral realities: does this investment decision align with this client's financial goals? Recent work shows that elicited attributes often diverge from observed actions, which raises concerns about whether existing standards reliably guide investors toward suitable choices or leave gaps in protection due to behavioral mismatches or inconsistent implementation.

Methodological applications

Clustering and similarity learning have been applied in a variety of contexts; however, an important and sometimes difficult question is choosing a clustering method that is best suited for a specific financial dataset. These evaluations include cluster cohesion and separation, similarity choice, and computational cost, where no single method dominates in all respects (Kou et al., 2014).

The choice of method can substantially affect insights, as illustrated in two recent studies using a financial wellness survey. One study focused on using the K-means approach to clustering (Phelps & Metzler, 2024), whereas a follow-up study extended this analysis using seven different distance metrics (Ghashti & Thompson, 2023). The first, using knowledge- and skill-based measures, identifies four clusters and emphasizes gaps between self-assessed and actual financial knowledge, which reveal risks of overconfidence and misperceived literacy. The second approach, using dissimilarity learning on mixed-type data, yields two clusters that differ primarily in terms of housing affordability, debt, budgeting, and retirement strategies. Although each method utilizes different similarities, they both

[6] https://www.finra.org/rules-guidance/rulebooks/finra-rules/2111
[7] https://www.ciro.ca/newsroom/publications/guidance-know-your-client-and-suitability-determination

demonstrate that financial wellness encompasses both cognitive misalignments and contextual stressors. As there are no "true" clusters, different similarity and clustering approaches offer unique insights into the same data that must be analyzed by finance-domain experts.

Behavioral finance, similarity learning and household finance

Identifying meaningful similarities between client trade and transaction behaviors help to group them into distinct investor types. Evidence from household finance shows that individuals with similar profiles on traditional financial data (such as income, education level, and self-assessed risk tolerance) often diverge in their actual saving, borrowing, and investment behaviors. Recent applications of clustering and similarity for investor classification (Thompson et al., 2021) address this gap by revealing misalignments between elicited investor profiles and observed decision-making.

Deep clustering has been used to identify heterogeneity in household financial behaviors that conventional demographics overlook (Hwang et al., 2023b). K-means clustering was used on literacy measures to reveal gaps between self-assessed knowledge and decision-making (Phelps & Metzler, 2024; Hwang et al., 2024). De Luca & Mehta (2023) studied investments in ESG among 5.5 million households using K-Modes clustering to identify five distinct investor types.

Determining similarity between clients allows financial institutions to better understand how their clients react to market stimuli and give better advice that aligns clients with their financial goals. Adaptive similarity learning methods for local information of financial information relevant to clustering allow methods to adapt to each unique financial dataset. Kernel metric learning has been used for clustering trade and transaction data of clients to determine trading behaviors under the guidance of financial advisors and during financial market crashes (Ghashti & Thompson, 2023, 2025a,b).

Pagliaro et al. (2021) analyzed over 1.5 million handwritten and drop-down notes written by financial advisors using a Customer Relationship Management system. They used the natural language processing method of Word2Vec to summarize textual information and achieve semantic similarity amongst a corpus of text, and then used logistic regression, decision trees, and gradient boosting to classify whether an investor would decide to move all assets to cash. Shiao et al. (2022) further used advisor notes, trade and transaction data, digital activity (such as log-ins, emails, and calls), and the VIX index with gradient boosting trees to predict whether investors would make similar impulsive trading decisions.

Kim et al. (2023) extended the healthcare analogy of "diagnosis and prescription" to the financial domain by introducing a healthcare diagnostic framework, Risk Information-adjusted Hierarchical Autoencoder (RI-HAE), that generates household financial risk scores. Health-state clustering groups households by similarity and stratifies them into risk-based clusters. In the context of future health costs, Tan & Mehta (2022) applied K-modes clustering to a health and retirement study and found that their categorizations yielded more accurate cost estimates for personalized retirement planning.

Clustering and Similarity for Other Asset Classes

This section reviews how clustering and similarity are applied beyond the core domains of equities, fixed income, funds, companies, and investors. We highlight use cases in commodities, currencies and foreign exchange, derivatives, and real estate (REITs). For each market, we start with practitioner applications and then summarize traditional and modern methods, organized by data type.

Commodities

Peer groups in commodities support diversification and hedging across energy, metals, and agriculture, help identify substitutes such as edible oils or battery metals, and interpret futures curves in terms of contango or backwardation relative to inventory and convenience yields (Gorton et al., 2013).

Traditional approaches. Early studies showed that commodities often move together despite differing fundamentals (Pindyck & Rotemberg, 1990). Clustering methods such as correlation networks and minimum spanning trees group assets into intuitive clusters like energy, precious metals, and agriculture, and track how these groups evolve over time (Lucey et al., 2011). Volatility-spillover similarity adds directionality, distinguishing commodities that transmit shocks from those that absorb them (Diebold & Yilmaz, 2012). Clustering can also be applied to futures curves: common contango or backwardation patterns reflect shared storage and carry dynamics, offering a basis for grouping commodities by market structure (Gorton et al., 2013).

Modern approaches (dynamic networks and multi-factor curves). Recent studies (Ma et al., 2021; Ouyang et al., 2022; Dahl et al., 2020; Szenderák, 2018) combine dynamic correlation with MSTs or connectedness indices to track evolving clusters and risk transmission. For specific complexes (such as oil), multi-factor models of the entire futures curve provide a compact representation of curve similarity across time and maturities (Cortazar & Naranjo, 2006). Results consistently show energy and precious metals form tight clusters, with agriculture being more diverse.

A complementary perspective is offered by Chen et al. (2021), who apply unsupervised learning to both the cross-section of commodities and the time dimension of trading days. Using k-means and hierarchical clustering on logarithmic returns (with manifold learning via Multi-dimensional Scaling and t-SNE for visualization), they recover economically interpretable peer groups across fuels, metals, and agriculturals. Crucially, they also cluster the *transpose* of the panel, grouping *trading days* by conditional volatility (GARCH-based), which reveals market regimes without imposing ad hoc crisis dates. For practitioners, this two-way clustering links instrument similarity (what trades together) with regime similarity (when markets behave alike), enabling regime-aware hedging and monitoring.

Currencies (Foreign Exchange)

Similarity in foreign exchange (FX) matters directly for risk management and trading. It guides hedge program design by grouping currencies into natural blocks, which reveals how shocks and interventions propagate across the FX network, and identifies safe-haven clusters such as the US dollar, Swiss franc, or Japanese yen that play a central role in risk parity and drawdown control.

Traditional approaches. Early studies show that exchange rates often transact at round numbers, with the strength of price clustering varying across currencies and trading platforms (Sopranzetti & Datar, 2002). Hierarchical clustering applied to return correlations or cointegration has long been used to identify regional blocks and key hub currencies (Kwapien et al., 2009; Wu & Zheng, 2013). (Scarlat et al., 2007) examined transitional economies and highlighted self-similar scaling properties in currency dynamics, suggesting deeper structural clustering even beyond correlation. (Wang et al., 2012) extended this work using DTW to measure similarity between entire return paths, producing minimum spanning trees that provide intuitive maps of the FX market structure.

Modern approaches. Recent work emphasizes the use of more complex similarity metrics and dynamic community detection. Mizuno et al. (2006) and Fenn et al. (2012) showed that correlation networks and community detection methods uncover currency groups that evolve over time in response to crises and interventions. Chakraborty et al. (2020) proposed similarity metrics based on the distribution of return fluctuations, recovering a hierarchical structure of the global FX market that goes beyond correlations. Fernandes et al. (2023) employed hierarchical clustering to compare West Texas Intermediate (WTI) oil, fiat currencies, and foreign exchange rates, identifying coherent clusters that spanned across asset classes, thereby highlighting the integration of energy and foreign exchange markets. Kremer et al. (2019) demonstrated that political and economic events alter clustering dynamics, changing the role of safe-haven currencies.

Derivatives

Similarity in derivatives helps in three main ways. It allows contracts to be grouped by their key risk exposures, such as delta, gamma, or vega (the options Greeks), reduces the complexity of large trading books by creating representative hedging clusters, and makes it possible to compare volatility surfaces and curve shapes across different underlyings to spot risks and relative-value opportunities.

Traditional Approaches. Option prices across strikes and maturities can be summarized by the implied volatility surface. This surface provides a natural way to measure similarity: contracts or markets with similar surface shapes behave alike. Empirical studies show that only a few factors explain most of the surface variation, so options can be clustered by their factor loadings or by the overall shape of their volatility surface (Derman & Kani, 1994).

Modern Approaches. In practice, traders already group options by sensitivities such as delta, gamma, and vega. Recent learning-based hedging frameworks formalize this by clustering contracts with similar exposure profiles, simplifying execution and valuation adjustments (Buehler et al., 2019).

In rates and commodities, contracts are naturally organized by the shape of their term structures. Classic studies showed that government bond yields can be explained by a few factors—level, slope, and curvature that capture most variation in returns (Litterman, 1991). This factor view underpins modern practice: bonds or swaps with similar loadings are treated as substitutes, and clustering by factor exposures is standard in fixed-income risk management. In commodities, futures curves are summarized in a similar way. Cortazar and Naranjo (Cortazar & Naranjo, 2006) show that two or three Gaussian factors describe oil futures, with peer contracts defined by comparable responses to contango or backwardation shifts.

Across asset classes, clustering instruments by factor loadings provides a compact and economically interpretable method for managing large derivative books.

Real Estate/REITs

In listed real estate and REITs, clustering by geography or property type helps define peers and manage risk across regions and sectors. For insurance-linked securities, similarity comes from contract features such as peril, trigger design, and expected loss, allowing contracts with comparable structures to be grouped for pricing, benchmarking, and portfolio construction.

Early studies (Gyourko & Keim, 1992; Ling & Naranjo, 1997; Goetzmann & Wachter, 1995) showed that real estate securities co-move with economic and financial factors and are closely tied to equities. This creates a natural basis for clustering: securities with similar factor exposures or geographic and sector footprints form peer groups that support benchmarking and risk analysis.

Economic Scenario Clustering and Similarity

Banks, insurers, and asset managers maintain large scenario libraries for regulatory stress tests such as Comprehensive Capital Analysis and Review (CCAR) and European Banking Authority (EBA), as well as for internal capital, Asset–Liability Management (ALM), and top-down risk exercises. Clustering and similarity make these sets usable by curating diverse but non-redundant scenarios, reducing Monte Carlo paths to tractable representatives, and assessing severity, novelty, and alignment with portfolio sensitivities such as credit, rates, housing, and labor. They also improve governance by ensuring scenario selection is auditable and reproducible Borio et al. (2014).

Data modalities. Scenario data appear in four forms: (i) **paths** (multivariate trajectories of GDP, unemployment, inflation, and so on); (ii) **trees** (multistage Monte Carlo or ESG outputs); (iii) **panels** (mixed-frequency macro indicators); and (iv) **narratives** (qualitative supervisory texts linked to quantitative paths).

Classical foundations *Scenario reduction* addresses the problem that Monte Carlo engines produce thousands of macro paths while regulators and managers need only a few. Forward and backward selection prune paths using probability distances such as Kantorovich or Wasserstein, preserving tail risks while cutting redundancy (Dupačová et al., 2003; Heitsch & Römisch, 2003). In clustering terms, reduction groups similar paths and selects prototypes, creating compressed yet decision-relevant libraries for ALM, energy, and solvency analysis.

Macro regime identification clusters the economy across time. Hamilton's Markov-switching model introduced latent states such as expansions and recessions (Hamilton, 1989), later extended with dynamic factor models to capture multiple indicators jointly (Camacho et al., 2018). These frameworks group periods with similar co-movements in GDP, unemployment, inflation, and financial variables, providing practitioners with structured, regime-based similarity for forecasting and stress testing.

Actuarial economic scenario generators simulate inflation, rates, equities, and real estate for insurers and pensions. Wilkie-style models capture long-run correlations, while Solvency II stresses calibration and transparency Moudiki & Planchet (2016); Ahlgrim et al. (2005). Clustering and similarity prune redundant scenarios to keep libraries of scenarios tractable, but regime coverage must be checked to avoid losing rare stress states. Once the reduction is applied, quite a few governance and compliance concerns can be resolved.

Modern methods Vanguard Research (2023) applied machine-learning clustering to classify capital market scenarios into five states (recession, recovery, expansion, slowdown, and high inflation), showing how unsupervised methods can build objective taxonomies for scenario analysis and portfolio design.

Trajectory similarity methods align cycles that differ in timing or intensity. Franses & Wiemann (2020) used DTW to recover comparable phases even when shifted in time. Augustyński & Laskoś-Grabowski (2018) applied DTW-based clustering to macro series, finding it more effective than correlation when recession timing varies across economies. Raihan (2017) showed that DTW improves U.S.

recession prediction, highlighting its early-warning value.

Stress-testing after the global financial crisis emphasized scenarios that are both severe and plausible. Glasserman et al. (2015) proposed empirical likelihood methods to generate extreme but consistent scenarios under supervisory limits. Aikman et al. (2024) introduced multi-scenario frameworks that cluster adverse mixes, demonstrating that multiple grouped scenarios capture vulnerabilities better than a single baseline.

Authorities now publish standardized paths. The Federal Reserve specifies GDP, unemployment, housing, and financial variables, enabling similarity checks against past vintages for novelty and redundancy[8] . The EBA provides EU-wide templates that can be clustered with earlier designs to ensure continuity while adapting to new risks[9].

Recent work explores generative and multimodal methods. Gustafsson & Jonsson (2023) applies Wasserstein GANs to generate synthetic stress-test scenarios, showing adversarial training can replicate and diversify macroeconomic paths. These advances mark a shift toward pipelines where clustering and similarity support reduction, evaluation, creation, and governance of scenarios.

Practical Tips for Practitioners

Clustering and similarity are powerful tools, but their value depends on design and governance. A few practical guidelines for financial modelers are below.

1. Define evaluation metrics before modeling. Even unsupervised projects need clear objectives, for example, for bonds, test substitution fidelity; for funds, category reproducibility; for investors, segment stability. Choosing Metrics right at the beginning anchors the pipeline in practitioner relevance.

2. Start with simple, interpretable methods. Use transparent baselines like correlation clustering, k-means, or hierarchical methods to set expectations. Only add advanced methods when incremental gains are clear.

3. Match distance metrics to data. Choose metrics suited to modality: Gower for mixed tabular data, DTW or correlation for time series, and so on. Avoid one-size-fits-all Euclidean distance.

4. Normalize and pre-process carefully. Scale and distribution differences distort similarity. Apply z-scores, log transforms, and so on to ensure results reflect the financial structure rather than units.

5. Use domain knowledge. Credit and liquidity matter for bonds, factors for funds, and risk tolerance for investors. Embed expert input through feature selection or must-link/cannot-link constraints to ensure operational relevance.

[8] https://www.federalreserve.gov/publications/files/2024-stress-test-scenarios-20240215.pdf
[9] https://www.eba.europa.eu/risk-and-data-analysis/risk-analysis/eu-wide-stress-testing

6. Apply supervised similarity when labels exist. Ratings, fund categories, or client segments allow supervised methods such as random forest proximities or Siamese networks. These produce task-specific similarity and richer interpretability.

7. Check stability across time. Markets shift with volatility and regimes. Use rolling windows, for example, and track stability metrics so practitioners know when peer sets are robust or regime-specific.

8. Document assumptions and design choices. Under SR 11-7, all features, preprocessing steps, metrics, and parameters must be recorded. Clear documentation ensures reproducibility, transparency, and regulatory acceptance.

9. Prioritize interpretability. Trust on the end results of clustering and similarity depends on explanations. Feature importances, SHAP values, or case-based reasoning (Yampolsky et al., 2024; Rosaler et al., 2024) show why entities are peers, which is critical in regulated contexts.

10. Plan for scale and production. Methods must scale to large universes. Use approximate nearest neighbors, distributed computing, and retraining pipelines to meet production needs.

Taken together, these principles make clustering and similarity systems interpretable, stable, and aligned with decision-making and governance.

Summary

Clustering and similarity learning have evolved from heuristic peer-grouping methods to become indispensable tools across asset classes. Traditional approaches relied on static classifications such as industry codes or style boxes, which offered only coarse approximations of economic relationships. In contrast, modern techniques enable adaptive, data-driven neighborhoods that better reflect actual risk, liquidity, and thematic exposures. By integrating clustering and similarity methods, practitioners can build flexible frameworks that structure complex financial data in economically meaningful and operationally useful ways, positioning these methods as core components of investment analytics, portfolio construction, and risk management systems. (Exhibits 6 to 12 summarize clustering and similarity methods applied in different asset classes and data modalities.)

A central lesson of the tutorial is that modeling choices must be closely aligned with the characteristics of the data and the intended use cases. Different modalities (tabular fundamentals, time-series returns, textual disclosures, and graphs) require distinct similarity techniques. Moreover, evaluation must move beyond abstract academic metrics to focus on task-specific criteria such as substitution fidelity, neighborhood stability, reproducibility, and utility in decision-making. These considerations ensure that similarity systems are transparent, auditable, and robust across market regimes, making them suitable for production deployment in regulated environments. Their applications are already broad, spanning bonds, funds, companies, and investors, and are rapidly extending into newer domains such as commodities, currencies, derivatives, and scenario generation.

Despite their promise, clustering and similarity methods face important practical challenges. Their outputs are often sensitive to initial conditions and hyperparameter choices, making results fragile and potentially unstable. Interpretability is another major hurdle: many high-performing models, especially deep learning approaches, operate as "black boxes," complicating accountability to clients and regulators. Additionally, similarity is inherently subjective and context-dependent, varying according to the analytical task, data modality, and market regime. These factors underscore that no single notion of similarity applies universally across all financial contexts.

The tutorial ultimately delivers a pragmatic message: clustering and similarity learning are no longer experimental techniques but essential, embedded components of modern financial workflows. To responsibly harness their full potential, practitioners must design robust, interpretable, and aligned models that align with fiduciary objectives. This requires thoughtful model selection, careful evaluation, and transparent governance. By addressing these issues directly, clustering and similarity frameworks can provide powerful tools for structuring financial information, enhancing decision-making, and meeting regulatory expectations.

Several domains remain under-explored but highly promising for clustering and similarity: commodities, derivatives, private markets, and ESG. In particular, similarities across private equity, private credit, private-to-public transactions, and so on are expected to become major focus areas for researchers in the near future. Advances in multimodal learning, regime-aware similarity, and open-set recognition will shape the next wave of applications.

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Appendix Online Tables

Exhibit 6: Clustering and similarity applications in fixed income across asset classes and data modalities

Asset Class	Data Modality	Method	Applications
Corporate Bonds	Tabular (ratings, spreads, fundamentals)	K-means Hierarchical Random Forest Proximity Metric Learning	Peer groups for valuation Illiquid bond substitutes Relative value screening
	Time series (returns, spreads)	Correlation Clustering Dynamic Time Warping Rolling-Winodw Clustering	Volatility regime detection Monitoring structural breaks Grouping by market behavior
	Graphs (ownership, issuer links)	Graph Embeddings (node2vec) Graph Neural Networks	Systemic risk & contagion analysis Finding substitutes via investor overlap
Municipal Bonds	Tabular (covenants, ratings)	Mixed-type Distances (Gower) k-prototypes, GB Proximity	Peer cohorts for surveillance Price check set construction Credit quality benchmarking
	Text (official statements, filings)	Text Embeddings + Clustering	Disclosure outlier detection ESG-relevant cohort analysis Monitoring communication consistency
	Graphs (issuer hierarchies, geo-links)	Community Detection, Spectral Clustering	Mapping regional exposure Monitoring systemic fiscal risks Identifying clusters for comparison
Securitized Credit (MBS/ABS)	Tabular (loan-level features)	Gaussian Mixtures Deep Metric Learning	Prepayment/default risk cohorts Borrower credit segmentation
	Time series (delinquency paths)	DTW Wasserstein Distances	Aligning stress test scenarios Benchmarking cohort performance Monitoring credit risk regimes
Housing / Collateral	Images & Spatial (photos, satellite)	Siamese Networks CNN Embeddings	Automated property comparables Collateral valuation & underwriting Regional stress testing

Exhibit 7: Clustering and similarity applications for mutual funds and ETFs across data modalities

Data Modality	**Method**	**Applications**
Time series (returns, volatility, flows)	Correlation Clustering K-shape DTW Contrastive Embeddings Regime-switching Models	Detecting investment styles Monitoring style drift Forming regime-aware peer groups Checking category reproducibility
Tabular (holdings, fees, AUM)	Cosine/Jaccard Similarity K-means Hierarchical Clustering Gower/k-prototypes Kernel Metric Learning	Identifying peer groups by holdings Detecting crowding & overlap risk Refining categories for due diligence Surveillance based on fund attributes
Text (prospectuses, reports)	TF-IDF Transformer Embeddings Supervised Similarity Models	Strategy & thematic classification ESG consistency analysis Verifying disclosure alignment Narrative anomaly detection
Graphs (fund–asset networks)	Community Detection Node Embeddings (DeepWalk, node2vec) Graph Neural Networks	Mapping systemic exposures Finding clusters of overlapping strategies Peer retrieval via network structure
Multimodal (returns + holdings + text)	Late/Early Fusion Cross-modal Transformers Joint Embeddings	Robust peer discovery Identifying innovative/hybrid strategies Allocation support with multiple sources

Exhibit 8: Clustering and Similarity Applications for Companies by Data Modality

Data Modality	**Method**	**Applications**
Tabular (fundamentals, ratios)	K-means Hierarchical Metric Learning Random Forest Proximities	Peer identification for valuation M&A target screening Benchmarking competitors
Time series (returns, volatility)	Correlation Clustering DTW Spectral Clustering Sequence Encoders	Thematic investing Market regime discovery Systemic risk diagnostics
Text (filings, news, patents)	TF-IDF + Clustering Transformer Embeddings LLM-based Similarity	Thematic grouping of firms Aligning strategy with disclosures Mapping product/technology overlaps
Graphs (supply chain, ownership)	Community Detection Node Embeddings (node2vec) Graph Neural Networks	Competitor mapping via relationships Tracing systemic risk channels Benchmarking governance/ownership
Correlation matrices (stock correlation networks)	Graph Embeddings GNNs Graph ML on Correlation Graphs	Peer discovery for portfolios Diversification analysis Systemic contagion monitoring
Multimodal (tabular + text + graphs)	Cross-modal Embeddings Multimodal Transformers Joint Representation Learning	Comprehensive peer discovery Mapping private to public comps Robustness across data sources

Exhibit 9: Clustering and Similarity Applications for Private Companies

Setting	Data Modality	Applications
Private-to-Private	Tabular (financials, registry data)	Benchmarking profitability Forming credit risk cohorts Identifying high-growth firms
	Text (websites, descriptions)	Startup ecosystem mapping Competitor identification Thematic grouping of SMEs
	Graphs (investors, partnerships)	Investor syndicate analysis Partnership discovery Competitive landscape mapping
	Images / Alternative Data (products, web traces)	Product & brand comparables Collateral & property valuation Digital footprint analysis
Private-to-Public	Tabular + Fundamentals (hybrid models)	Relative valuation vs. public peers Sector benchmarking Building interpretable comparables
	Text + Financials (multimodal)	Aligning private firms with public disclosures Refining hybrid comparables
	Graphs (shared investors, supply chains)	Ecosystem mapping with public firms M&A target discovery Tracing systemic interdependencies
	Time series (synthetic proxies)	Benchmarking volatility with proxies Stress testing & scenario analysis

Exhibit 10: Clustering and similarity applications for investors across data modalities

Data Modality	Method	Simplified Applications
Tabular (demographics, portfolio)	K-means Hierarchical Gower Distance K-prototypes	Client segmentation for product design Suitability checks Targeted advisory services
Transaction Histories (trades, flows)	DTW Correlation Clustering Contrastive Embeddings	Detecting herding behavior Monitoring liquidity risk Tailoring engagement strategies
Risk & Behavioral Signals (questionnaires, sentiment)	Latent Class Analysis Mixture Models Supervised Similarity Learning	Risk tolerance profiling Suitability & compliance monitoring Personalized allocation strategies
Text (surveys, advisor notes)	Topic Modeling, Transformer Embeddings	Thematic segmentation (goals, ESG) Advisor support tools Product recommendation
Graphs (social, referral networks)	Community Detection Spectral Clustering GNNs	Detecting peer influence Identifying contagion pathways Mapping behavioral communities
Multimodal (portfolios + flows + text)	Cross-modal Embeddings Multimodal Transformers	Comprehensive client segmentation Robo-advisor personalization Stress-test propagation analysis

Exhibit 11: Clustering and Similarity Applications in Other Asset Classes

Asset Class	**Method / Modality**	**Simplified Applications**
Commodities	Time-series Clustering (correlations, DTW, regimes)	Diversification & hedging strategies Identifying contagion channels Analyzing co-movements
FX (Currencies)	Correlation-based Clustering Hierarchical Clustering	Identifying currency blocs Designing carry trade baskets Managing FX exposure
Derivatives (Options, CDS)	Manifold Learning Volatility Surface Similarity Clustering CDS Spreads	Market surveillance & anomaly detection Relative pricing of options Systemic risk & counterparty monitoring
Cryptocurrencies	Network Clustering, Time-series Correlation Clustering, Transaction Graph Embeddings	Identifying stable coin families Monitoring contagion channels Assessing systemic risk in digital assets
Real Assets / Alternatives	Image & Spatial Embeddings, Feature-based Clustering	Collateral valuation & comparables Geographic concentration analysis Peer benchmarking & risk monitoring)

Exhibit 12: Summary of data modalities, methods, and applications for economic scenario clustering and similarity

Data Modality	**Method**	**Applications**
Paths (macro trajectories)	Dynamic Time Warping (DTW), Trajectory Clustering, Correlation-based Clustering	Macro regime classification Cross-country cycle alignment Early warning systems for recessions
Scenario Trees (multistage Monte Carlo)	Wasserstein / Nested Distance, Entropic Regularization	Scenario reduction for ALM Risk management optimization Preserving key decision features
Panels (mixed-frequency indicators)	Factor-based Clustering, Regime-switching Models (e.g., Markov Switching)	Probabilistic regime identification Macroeconomic forecasting Stress-test design
Narratives (supervisory texts)	Topic Modeling, Text Embeddings, Bayesian Networks	Aligning narratives with quantitative paths Plausibility checks & governance Supervisory scenario analysis
Stress Scenarios (macro-financial variables)	Clustering of Severity Metrics, Empirical Likelihood Methods	Systematic supervisory scenario design Plausibility benchmarking Ensuring diversity of adverse cases
Multimodal (quantitative + qualitative)	Cross-modal Embeddings, Multimodal Transformers, GANs for scenario generation	Generating diverse & plausible scenarios Aligning designs with historical data Predicting systemic vulnerabilities